

Barter Exchange with Asymmetric Item Valuations

Juan Luque
University of Maryland
College Park, MD, USA
jluque@umd.edu

Sharmila Duppala
University of Maryland
College Park, MD, USA
sduppala@umd.edu

Michael Curry
University of Illinois Chicago
Chicago, IL, USA

John P. Dickerson
Mozilla AI
Seattle, Washington, USA
john@mozilla.ai

Aravind Srinivasan
University of Maryland
College Park, MD, USA
asriniv1@umd.edu

Abstract

Agents enter barter exchanges to swap items they have for items they want. We study Barter Exchange with Asymmetric Valuations (BAV), a centralized barter exchange where each agent has an individual valuation over items. Given a reallocation of items, let an agent's profit be their received value minus their value given away, according to said agent's valuation of items. The goal of the clearinghouse (the party facilitating the exchange) is to output a reallocation of items that maximizes welfare (sum of agent profits) subject to each agent receiving non-negative profit.

BAV is a natural extension of the barter exchange model studied in (Luque et al., WWW '24), which assumes agents necessarily agree on item values. We prove BAV is strongly NP-hard and design a polynomial-time randomized algorithm with optimal expected welfare, non-negative expected profit for every agent, and that, with probability 1, every agent is at most the value of two items away from positive profit. Since valuations may be private and revealed by self-interested agents, we also study BAV from a mechanism design lens.

CCS Concepts

• **Theory of computation** → **Algorithmic game theory and mechanism design**; **Rounding techniques**; • **Applied computing** → Online auctions.

Keywords

Barter Exchange, Mechanism Design, Approximation Algorithms, Randomized Algorithms, Combinatorial Optimization, Linear Program Rounding

ACM Reference Format:

Juan Luque, Sharmila Duppala, Michael Curry, John P. Dickerson, and Aravind Srinivasan. 2026. Barter Exchange with Asymmetric Item Valuations. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3774904.3792349>



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WWW '26, Dubai, United Arab Emirates

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ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792349>

1 Introduction

From time immemorial, barter has existed as a means of trade for goods and services. Revolutionary as the invention of money has been, barter exchanges persist. Bartering is simple and is still preferable in many cases; in some cases it is the only legal means of exchange. Yet, as originally conceptualized, bartering is plagued by the steep requisite of a “double coincidence of wants” [10]. The proliferation of social media and e-commerce has streamlined the exchange of goods and carved a space for new barter exchanges by vastly increasing the pools of eligible parties to trade with. For example, Facebook and Instagram are massive markets where users may buy and sell their goods to one another. On Reddit, many communities exist as barter exchanges where users gather to trade board games, video games, video game items, and trading cards. To name a few, the subreddit GAMESWAP¹ (61,000 members) and Facebook group BOARDGAMEEXCHANGE² (51,000 members) are communities where exchange video games and board games. The existence of these online communities is proof that, even when items can be sold, an exchange might be more convenient.

In a *centralized* barter exchange, a platform, which we will refer to as the *clearinghouse*, facilitates the exchange. The clearinghouse, aided by the access to massive markets via internet communities, can easily overcome the “double coincidence of wants” by coordinating “many-to-many” swaps where one agent can give and receive many items to and from many agents. In contrast, in a *decentralized* exchange agents coordinate swaps independently with each other. The greater level of coordination possible in a centralized exchange unlocks reallocations that maximize its participants' welfare and minimizes their required effort to obtain suitable swaps.

Kidney exchange is a canonical example of barter exchange because the sale of organs is outlawed in all countries except for Iran. For patients in need of a transplant, barter is a necessity. In the Kidney Exchange problem, patient-donor pairs, consisting of a patient in need of a donor organ and a willing but incompatible donor, enter a centralized exchange with the goal of being matched with a compatible donor in exchange for their donor's organ. Aided by the power of modern computation and algorithms, the clearinghouse can compute complex reallocations of donor organs in cycles, whose length is often limited to 3 or 4 but can be as large as 10 ([8]), or in chains [14]. Kidney Exchange Programs [1, 3] have had great success in that they greatly increase the number of transplants

¹www.reddit.com/r/Gameswap

²<https://www.facebook.com/groups/boardgameexchange>

occurring compared to a decentralized exchange. Computing optimal allocations is often NP-hard in practice, so solutions include heuristics and approximations [4, 7, 14].

We study a centralized barter exchange problem where agents enter with lists of items owned and desired, as well as valuation profiles over said items. The goal is to compute a (re)allocation of items that maximizes welfare and where every agent receives at least as much value as they give away. We call this problem *Barter Exchange with Asymmetric Item Valuations* (BAV). If all agents agree on the item values, this is called *Barter Exchange with Symmetric Item Valuations* (BSV) problem, studied in Luque et al. [12]. Although it is often possible to find a reasonable consensus valuation, say, from fair market value, such assumptions are not always reasonable. Moreover, granular preference expression over allows for more efficient item allocations in terms of maximizing the happiness of participating agents. For example, Prendergast [15] studies food banks bidding for more efficient allocations where the increased efficiency stems from the asymmetry in the needs and preferences of different food banks.

We first consider the problem assuming knowledge of agent valuations. Our algorithmic results follow the framework of Saha and Srinivasan [16] where the combinatorial optimization problem of interest is first cast as a linear program (LP), and whose solution is subsequently judiciously rounded via a random walk inside the polytope spanned by the LP constraints, while interspersing a careful relaxation of some problem constraints. It combines iterative rounding [9, 11, 17] and (randomized) dependent rounding [2, 5, 18] techniques.

Since each agent's valuation function is possibly private, we also study the problem from a mechanism design perspective, where agents may misreport their values. The goal of the mechanism designer is to select a barter that achieves some goals related to quality of outcome and incentives produced for the agents. While in the algorithmic problem we allow barter only, in the mechanism design problem we consider that monetary transfers between the center and agents may be possible. Loosely defined, the desiderata we consider in this paper include welfare maximization (the reallocation maximizes happiness), individual rationality (every agent is weakly better off participating in the exchange than not), and incentive compatibility (agents maximize their utility by reporting truthfully).

2 Preliminaries

A set of linear constraints C , for example “ $Ax \leq b$ ”, defines a polytope $\mathcal{P} = \{x \mid Ax \leq b\}$.

Definition 1 (Extreme point). x is an extreme point of a polytope $\mathcal{P}$ if for all $y \neq 0$, either $x + y \notin \mathcal{P}$ or $x - y \notin \mathcal{P}$.

If x is not an extreme point of $\mathcal{P}$, we can apply the subroutine $\text{RANDOMMOVE}(x, C)$ from [16]. $\text{RANDOMMOVE}(x, C)$ returns x' where,

$$\forall i, \mathbb{E}[x'_i] = x_i \quad (1)$$

and $x' \in \mathcal{P}$, keeps all previous constraints tight and makes an additional constraint tight. That is,

$$\{i \mid (Ax)_i = b_i\} \subset \{i \mid (Ax')_i = b_i\}. \quad (2)$$

We call constraints in C *active*. If a constraint holds with equality it is said to be *tight*. We use multiple applications of $\text{RANDOMMOVE}(C)$ to judiciously round a vector $x \in [0, 1]^d$ to $X \in \{0, 1\}^d$ while ensuring active constraints C are preserved. A constraint c dropped from C is said to be *inactive*. By dropping constraints carefully, we ensure two things. First, in any given iteration, x is not an extreme point of C so RANDOMMOVE can be applied. Second, we bound dropped constraints' worst-case violations in X . An element x_i is said to be *floating* if $0 < x_i < 1$ and it is said to be *settled* if $x_i \in \{0, 1\}$. The rounding algorithm will terminate when all elements of x are settled and return X .

3 Problem formulation

A BAV instance consists of n agents and a universe $\mathcal{I}$ of m items. Agent i has items $H_i \subseteq \mathcal{I}$ and wants items $W_i \subseteq \mathcal{I}$. Moreover, each agent has a valuation function $v_i: H_i \cup W_i \rightarrow \mathbb{R}^+$ over said items. After reallocating items, each agent has items $A_i \subseteq H_i \cup W_i$. For $S \subseteq \mathcal{I}$, let $v_i(S) := \sum_{j \in S} v_i(j)$; therefore, each agent has utility (profit) $\mu_i(A_i) := v_i(A_i) - v_i(H_i)$.

The task posed in this Barter Exchange problem is finding a (valid) re-allocation $A_1, \dots, A_n$ maximizing the welfare $\sum_{i=1}^n \mu_i(A_i)$ subject to $\mu_i(A_i) \geq \pi_i$ where π_i are profit demand for agent i . A BSV instance is thus specified by the tuple $(\mathcal{I}, \{(H_i, W_i, v_i, \pi_i)\}_{i=1}^n)$.

3.1 Equivalent graph matching formulation

The allocation $A_1, \dots, A_n$ above needs to be valid. That is, the final allocation must be possible by moving items one at a time where agents can only give away items they own and receive items that they want; items can neither be created nor destroyed. We introduce a graph matching variant to ensure the allocation is valid and profit constraints are respected.

Define the bipartite graph (L, R, E) where $L := \dot{\cup}_{i \in [n]} L_i$ and $R := \dot{\cup}_{i \in [n]} R_i$, and L_i (R_i) has a vertex for each item in H_i (W_i). Formally, $L_i := \{\ell_{i,j} \mid j \in H_i\}$ and $R_i := \{r_{i,j} \mid j \in W_i\}$. We extend an agent i 's valuation over item j to its corresponding vertices: so $v_i(\ell_{i,j}) = v_i(r_{i,j}) = v_i(j)$. Finally, an edge is drawn between left and right vertices corresponding to the same item: $E := \{(\ell_{i,j}, r_{i',j}) \mid i, i' \in [n]\}$. Let $X \in \{0, 1\}^{|E|}$ define the graph matching $M = \{e \in E \mid X_e = 1\}$. M is constrained to be a matching via the usual edge selection and vertex degree constraints:

$$X_e \in \{0, 1\}, \quad \forall e \in E, \quad (3)$$

and, for $X(a) := \sum_{e=(a,b) \in E} X_e$,

$$X(a) \leq 1, \quad \forall a \in L \cup R. \quad (4)$$

$(\ell_{i,j}, r_{i',j}) \in M$ means “agent i gives item j to agent i' ,” (or, equivalently, “agent i' receives item j from agent i ”). This equivalency ensures that the reallocation neither creates nor destroys items. In other words, $X(\ell_{i,j}) = 1$ ($X(r_{i,j}) = 1$) if and only if agent i gives away (receives) item j .

Finally, agent i has utility (profit)

$$\mu_i(X) = \sum_{a \in R_i} v_i(a)X(a) - \sum_{b \in L_i} v_i(b)X(b) \quad (5)$$

and its profit demand is enforced with

$$\mu_i(X) \geq \pi_i, \quad \forall i \in [n]. \quad (6)$$

A reallocation X is said to be *valid* if it satisfies (3) and (4). The goal of BAV is to find a valid reallocation X subject to (6) and maximizing

$$\sum_{i=1}^n \mu_i(X). \quad (7)$$

Comparison to BSV. The main difference between these two problems is that BSV requires $v_i(j) = v_{i'}(j)$ for all $i \neq i'$ and $j \in \mathcal{I}$. Therefore, BAV allows edges between vertices with different values but BSV does not. The objectives are also different: BAV maximizes (7) and BSV maximizes the value exchanging hands (i.e., $\sum_{a \in R} v_i(a)x(a)$).

4 BAV Approximation Algorithm

Finding any feasible non-trivial reallocation x as described in Section 3.1 is NP-hard. We pivot strategies by relaxing (3) to

$$0 \leq x_e \leq 1, \quad \forall e \in E \quad (8)$$

and finding a *fractional* reallocation $x \in [0, 1]^{|E|}$ in polynomial time by writing and solving the LP $\max_x \sum_{i=1}^n \mu_i(x)$ subject to analogous constraints (8), (4), and (6). We refer to this LP as LP-Barter. While a fractional reallocation will not suffice for indivisible items, the following theorem shows how RALLOC, Algorithm 1, lets us efficiently round $x \in [0, 1]^{|E|}$ to $X \in \{0, 1\}^{|E|}$ while enjoying near-optimality in welfare and agent profit guarantees.

Algorithm 1 RALLOC($x, D \cup W \cup E'$)

Require: $x \in [0, 1]^{|E|}$ is a fractional allocation with sets D, W , and E' of active degree, profit, and edge constraints; i.e., (4), (6), and (8), respectively. $x \in \mathcal{P}(D \cup W \cup E')$.

Ensure: $X \in \{0, 1\}^{|E|}$ is an integral reallocation

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1:  $D \leftarrow$  set of active degree constraints, (4)
2:  $W \leftarrow$  set of active profit constraints, (6)
3:  $E' \leftarrow$  set of active edge constraints, (8)
4: while  $0 < x_e < 1$  for some  $e \in E'$  do
5:   for  $w \in W$  do
6:      $V(w) \leftarrow$  vertices in constraint  $w$ 
7:     if  $|\{a \in V(w) \mid 0 < x(a) < 1\}| \leq 2$  then
8:        $W \leftarrow W - \{w\}$ 
9:     if  $x_e \in \{0, 1\}$  for some  $e \in E'$  then
10:       $E' \leftarrow E' - \{e\}$ 
11:      Fix  $x_e \in \{0, 1\}$  and update  $x, D$ , and  $W$ 
12:     else  $\triangleright x$  is not an extreme point of  $\mathcal{P}(D \cup W \cup E')$ 
13:        $x \leftarrow \text{RANDOMMOVE}(x, D \cup W \cup E')$ 
14:  $X \leftarrow x$ 
15: return  $X$ 
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Let $v_i^* := \max_{j \in H_i \cup W_i} v_i(j)$.

THEOREM 1. Take a set of degree, value, and edge constraints D, W , and E' , and a fractional reallocation $x^* \in \mathcal{P}(D \cup W \cup E'')$. RALLOC($x^*, D \cup W \cup E''$) is a polynomial-time randomized algorithm that rounds x^* to a valid reallocation $X \in \{0, 1\}^{|E|}$ satisfying

- (1) $\forall i, \mathbb{E}[\mu_i(X)] \geq \pi_i$,
- (2) $\forall i, \mu_i(X) > \pi_i - 2v_i^*$ with probability 1, and
- (3) $\mathbb{E}[\sum_{i \in [n]} \mu_i(X)] = \mathbb{E}[\sum_{i \in [n]} \mu_i(x^*)]$.

PROOF. First observe that since the degree constraints remain active, X is valid. RALLOC terminates in polynomial time since the while loop executes polynomially-many times per Lemma 2, and each iteration is itself polynomial time.

By induction and the marginal property of RANDOMMOVE, $\mathbb{E}[X_e] = x_e$ for all $e \in E$. Therefore, by the linearity of expectation

$$\begin{aligned} \mathbb{E}[\mu_i(X)] &= \sum_{a \in R_i} v(a) \mathbb{E}[X_e] - \sum_{b \in L_i} v(b) \mathbb{E}[X_e] \\ &= \sum_{a \in R_i} v(a) x_e - \sum_{b \in L_i} v(b) x_e \\ &= \mu_i(x) \geq \pi_i. \end{aligned}$$

A similar application of the linearity of expectation proves the third property.

For the second property, fix a profit constraint w belonging to some agent i . If w remains active for the entirety of the algorithm, then RANDOMMOVE will not violate it and we are done. Otherwise, when w becomes inactive it has floating vertices

$$F = \{a' \in V(w) \mid 0 < x(a') < 1\}$$

with $0 \leq |F| \leq 2$. Let x' be x right after w becomes inactive and before the next call to RANDOMMOVE. Then $\mu_i(x') \geq \pi_i$ and

$$\begin{aligned} \mu_i(x') &= \sum_{a \in R_i - F} v_i(a)x'(a) - \sum_{b \in L_i - F} v_i(b)x'(b) \\ &\quad + \sum_{a \in R_i \cap F} v_i(a)x'(a) - \sum_{b \in L_i \cap F} v_i(b)x'(b) \end{aligned}$$

Since degree constraints never become inactive, RANDOMMOVE ensures $X(a) = x'(a)$ for all $a \in (L_i \cup R_i) - F$. Therefore,

$$\begin{aligned} \mu_i(X) - \mu_i(x') &= \sum_{a \in R_i \cap F} v_i(a)(X(a) - x'(a)) \\ &\quad - \sum_{b \in L_i \cap F} v_i(b)(X(b) - x'(b)). \quad (9) \end{aligned}$$

Intuitively, the right hand side is minimized when $|F| = 2$ and the rounding of both $X(s)$ for $s \in F$ correspond to agent i giving away (or failing to receive) its most expensive items. For example, if $|F| = |F \cap L_i| = 2$, $|F \cap R_i| = 0$, and $X(b) = 1$ for $b \in L_i \cap F$, then

$$\mu_i(X) - \mu_i(x') > -2v_i^*. \quad (10)$$

The inequality is strict because $0 < x'(b) < 1$. The same argument gives (10) when $|F \cap L_i| = |F \cap R_i| = 1$ or $|F \cap R_i| = 2$. By definition of x' , this is the first iteration where (6) for agent i could be violated. Therefore, $\mu_i(x') \geq \pi_i$, completing the proof. $\square$

To prove the correctness of RALLOC, it suffices to prove that its calls to RANDOMMOVE do not fail and that the while-loop runs a polynomial number of times. These two claims are proved in Lemmas 1 and 2, respectively.

LEMMA 1. If $\forall e \in E, 0 < x_e < 1$ and x is not an extreme point of its active constraints polytope, then there exists a value constraint with at most two floating vertices.

PROOF. Our proof proceeds via a typical iterative rounding argument. Our iterative rounding algorithm consists of dropping all value constraints (i.e., (6)) involving two or fewer floating vertices;

that is, vertices v with $0 < x(v) < 1$. As usual, if x_e equals 0 or 1, we fix it and adjust the graph and constraints accordingly.

Let D' and W' be the set of tight degree and value constraints making up a maximal number of linearly independent constraints. E is the set of edges, for which we have assumed $0 < x_e < 1$ for all $e \in E$. For $w \in W'$, let $V(w) \subseteq V$ be the set of vertices appearing in constraint w . Similarly, let $V(D') \subseteq V$ be the set vertices with tight degree constraints. Let d_v denote the degree of v . By a usual Rank Lemma [11] we know that $|E| = |D'| + |W'|$.

We will use a token counting argument to show $|E| > |D'| + |W'|$. Let $(u, v) \in E$ pay $1/2$ token to each of u and v . Therefore, $|E|$ tokens are paid out.

The tight constraints collect tokens from vertices as follows. Every $d \in D'$ is the tight degree constraint of some vertex v , so let d collect all the tokens v gets. Since $\deg(v) \geq 2$, d collects ≥ 1 tokens.

Similarly, a tight value constraint $w \in W'$ collects all tokens from $v \in V(w) - V(D')$. Since $v \notin V(D')$ and $\forall w' \in W' - \{w\}$, $V(w) \cap V(w') = \emptyset$, its tokens are indeed unclaimed thus far. We dropped all constraints in W' with two or fewer floating vertices; therefore w has ≥ 3 floating vertices. Therefore, $|V(w) - V(D')| \geq 3$ and w collects ≥ 1.5 tokens.

Finally, we have $|E| \geq |D'| + 1.5|W'|$, a contradiction assuming $W' \neq \emptyset$. We conclude the argument by remarking that $W' = \emptyset$ implies we are exactly looking at bipartite graph matching. In that case, we can complete the proof in the usual way: $V = V(D')$ and there is a cycle in the graph, contradicting the linear independence of constraints D' . $\square$

LEMMA 2. *RALLOC terminates in $O(mn^2)$ iterations.*

PROOF. By construction RALLOC, drops profit constraints with two or fewer floating vertices. Therefore, by the contrapositive of Lemma 1, every iteration either x_e just became 0 or 1 or x is not an extreme point of $\mathcal{P}(C)$. In either case, either one variable x_e just became settled or RANDMOVE can be applied to x resulting in x' where either a newly settled x_e or a newly tight active constraint where once a constraint becomes tight it remains tight.

For each $\ell_{i,j} \in L_i$,

$$\deg(\ell_{i,j}) = |\{i \in [n] \mid j \in H_i \cup W_i\}| \leq n,$$

so $|E| = \sum_{a \in L} \deg(a) \leq mn^2$. There are $2mn$ degree constraints and n profit constraints. Therefore, the algorithm terminates in $O(mn^2)$ iterations. $\square$

By starting from an optimal solution x^* of LP-Barter and applying Theorem 1, the next result is immediate.

THEOREM 2. *Assuming LP-Barter is feasible for the given profit demands π_i , we can find a reallocation X with objective optimal in expectation and where, for all i , $\mathbb{E}[\mu_i] \geq \pi_i$ and $\mu_i(X) > \pi_i - 2\vartheta_i^*$ with probability 1.*

Consider the “trivial” reallocation $x = 0$:

REMARK 1. *LP-Barter is feasible if $\pi_i = 0$ for all i .*

5 Hardness of BAV and BSV

For BSV, Luque et al. [12] prove that even deciding whether a non-trivial reallocation (i.e. $x \neq 0$) exists is weakly NP-hard. We complement this result with a strong NP-hardness result for the decision version. Let the decision version of BSV, SymBarter, ask “is there a reallocation moving at least k units of value where each agent gives and receives equal value?”

THEOREM 3. *SymBarter is strongly NP-hard.*

PROOF. Reduce from 3PART [6]. Take the instance $a_1, \dots, a_{3n} \in \mathbb{N}$ with $\sum_i a_i = Tn$. The corresponding SymBarter instance is given by $n + 1$ agents and $I = [4n]$. Then $H_1 = \{1, \dots, 3n\}$, each with $v_j = a_j$ for $j \in H_1$, and $W_1 = \{3n + 1, \dots, 4n\}$. For the remaining agents with $i \geq 2$, $H_i = \{3n - 1 + i\}$, $v_{3n-1+i} = T$, and $W_i = H_1$.

A moment’s reflection shows that there exists a valid 3PART iff there exists a valid SymBarter allocation of value at least $2nT$. For the allocation to have value $2nT$, all items must have been moved. Particularly, each of agents 2 through $n + 1$ must have given away their item of value T and received items of total value T in return. Let $f(j)$ be the agent receiving item j . Then $S_2, \dots, S_{n+1}$ where $S_i = \{a_j \mid f(j) = i\}$ defines a 3PART since $\sum_{j:f(j)=i} a_j = \sum_{j:f(j)=i} v_j = T$.

For the other direction, indeed a 3PART will define a function $g(j) = i$ saying a_j is placed in triple S_i . This precisely defines a SymBarter allocation where agent 1 gives item j to agent $g(j) + 1$. Indeed agent $i \geq 2$ gives away value T to agent 1 and receives items $\{j \mid g(j) + 1 = i\}$ for a total value of $T = \sum_{j:g(j)+1=i} v_j = \sum_{j:g(j)+1=i} a_j$. Agent 1 gives away and receives back nT units of value. Summing over values given away by each agent we arrive at an allocation moving $2nT$ units of value where each agent gives away and receives equal value. $\square$

The decision version of BAV, AsymBarter, asks “does there exist a reallocation X where $\sum_{i \in [n]} \mu_i(X) \geq k$ with $\mu_i(X) \geq \pi_i$ for all i .”

THEOREM 4. *AsymBarter is strongly NP-hard even if $\pi_i = 0$ for all i .*

PROOF. Reduce from 3PART variant with $a_j \in (T/4, T/2)$ for all j . Like before, define H_i and W_i for $1 \leq i \leq n + 1$ the same way over the same set I of $4n$ items. Set $\pi_i = 0$ for all i . All swaps occur between agent 1 and another agent $i \geq 2$. For all $i \geq 2$, leave the valuations the same: $v_i(j) = a_j$ for $1 \leq j \leq 3n$ and $v_i(j) = T$ for $3n + 1 \leq j \leq 4n$. For the first agent, let $v_1(j) = v_i(j)$ for $j \in H_1$ and $v_1(j) = 2T$ for $j \in W_1$. Every time agent 1 receives $j \in W_1$, the swap creates $v_i(j) - v_1(j) = 2T - T = T$ units of value. Every other item creates zero value since they are valued the same by the giving and receiving agents.

Therefore, there is a reallocation creating nT units of value if and only if there is a valid reallocation where agent 1 receives every item from agents $2, \dots, n + 1$. Call said reallocation X . In X , every agent i , for $i \geq 2$, must receive $\geq T$ units of value else $\mu_i(X) < 0$. Since all agents agree on the values of $j \in H_1$, there is a total value of nT for $j \in H_1$, and there are n agents each expecting T units of value, it must be that every agent receives exactly T units of value in X . In conclusion, there is a valid reallocation if and only if each agent i for $i \geq 2$ receives exactly T units of value. For $i \geq 2$, let $S_i = \{j_{i1}, j_{i2}, j_{i3}\} \subseteq W_i$ be the set of items received by agent i ; we

may assume $|S_i| = 3$ since $v_i(j) = a_j \in (T/4, T/2)$. Then $\mu_i \geq 0$ if and only if $S'_i = \{a_j \mid j \in S_i\}$ is a 3-Partition. In conclusion, there is a valid reallocation X creating nT units of value if and only if there is a valid 3-partition $S'_2, \dots, S'_{n+1}$. $\square$

6 Mechanism Design for BAV

In this section, we explore mechanisms for BAV. Toward the end of satisfying desirable properties such as incentive compatibility, we will introduce *payments*. The mechanisms studied will consist of a reallocation X and a set of payments $P_i(X)$ either paid to or collected from each agent i . These payments could be cash, IOUs, “funny money,” as long as agent utilities are plausibly quasilinear in the payments. Payments serve to bring about desirable properties in the mechanism. In the same spirit as in Theorem 2, we ideally want to ensure that the mechanism’s output reallocation X leaves agents happy even before any payments are doled out. As a secondary purpose, in the case (due to the indivisibility of items) that an agent loses value in the reallocation, money is a better-than-nothing alternative that helps make agents whole. Without caring for reallocations that make agents happy prior to payments, our Barter Exchange problem degenerates into a double auction equivalent to agents buying and selling items among themselves.

This model captures a setting where agents want to have items, either because they enjoy collecting them, or because the agent has urgent needs and money is only valuable as a bargaining chip for the desired goods. In Feeding America, food banks bid on food donations with specialized currency to receive better allocations [15]. Suppose instead that local food banks enter into a food exchange with one another where the asymmetric valuations reflect the different food banks’ needs, which may change over time alongside their clientele, supplies, natural disasters, and so on. In this food exchange, we could introduce specialized currency to elicit a mechanism with desirable properties. However, funny money is helpful as a leverage for more food in a future exchange. It is unacceptable if a food bank enters the exchange and “sells” off all its supplies in exchange for the specialized currency. The food bank must maintain a healthy supply to tend to its customers, which our algorithm ensures.

We motivate our exploration of mechanism design by noting agents may misreport to improve their utility. Let Alice and Bob each have a single item to trade with one another. They both value their own and the other’s item at 1 and 2 units of value, respectively. In the welfare maximizing (randomized) reallocation both items are swapped with probability 1. However, if Alice lies and claims that Bob’s item is instead worth only 0.9 units of value to her, she will give away her item with some probability $p < 1$, thereby increasing her utility at the expense of Bob. Similarly simple examples show that selfish incentives can come at an unacceptably high cost of welfare. With this simple example in mind, we turn to the question of IC barter exchange mechanisms. We center our search for IC mechanisms around Vickrey-Clarke-Groves mechanisms.

Agent s will report valuation v'_s . Fix the reported valuations v_{-s} of all other agents. When reporting v'_s agent s has *apparent* utility

$$\mu'_s(X) = \sum_{a \in R} X(a)v'_s(a) - \sum_{b \in R} X(b)v'_s(b) \quad (11)$$

as in (5) but with v'_s in place of v_s . Let

$$S = \{x \in [0, 1]^{|E|} \mid \text{Eq. (4) and } \forall i, \mu_i(x) \geq 0\}. \quad (12)$$

Then

$$x^* = \arg \max_{x \in S} \sum_{i=1}^n \mu_i(x) \quad (13)$$

is the optimal fractional reallocation under truthful report $v'_s = v_s$, and

$$x^*(v'_s) = \arg \max_{x \in S} \mu'_s(x) + \sum_{i=1, i \neq s}^n \mu_i(x) \quad (14)$$

is the optimal fractional allocation under a reported v'_s . Fractional reallocations are rounded using RALLOC, using a set W of initially active profit constraints specified for each mechanism. Thus, rounding with RALLOC yields reallocations X and $X(v'_s)$.

After reallocation and payment, s has utility

$$u_s(X(v'_s), P_s(X(v'_s))) = \mu_s(X(v'_s)) + P_s(X(v'_s)). \quad (15)$$

We write P_i instead of $P_i(X)$ when X is clear from the context.

A mechanism is said to be *interim IR* if $\mathbb{E}[u_i(X)] \geq 0$ for all i , and *ex-post IR* if $u_i(X) \geq 0$ with probability 1. Earlier we motivated reallocations X where agents are happy with the item reallocation even before payments. To this end we introduce a notion of *full individual rationality*. A mechanism is said to be *fully IR* (or to satisfy full IR) if, before payments, the reallocation is interim IR and ex-post IR up to two items, and it remains ex-post IR after payments. Precisely, the mechanism (X, P) is said to be fully IR if

- (1) $\mathbb{E}[\mu_i(X)] \geq 0$;
- (2) $\mu_i(X) > -2v_i^*$ with probability 1, and
- (3) $u_i(X, P) \geq 0$ with probability 1.

This is a notion of IR that additionally selects for reallocations where agents are (sufficiently) happy independent of payments. It is stronger in the sense that

$$\text{fully IR} \implies \text{ex-post IR} \implies \text{interim IR}.$$

A mechanism is welfare maximizing (WM) if it outputs a reallocation X with

$$\mathbb{E} \left[\sum_{i=1}^n \mu_i(X) \right] \geq \sum_{i=1}^n \mu_i(x^*). \quad (16)$$

That is, X has expected welfare at least that of the optimal fractional reallocation. A mechanism is said to be *weakly budget balanced* (WBB) if $\mathbb{E} \left[\sum_{i=1}^n P_i \right] \leq 0$. This means the clearinghouse must bring at most an expected 0 units of money (from their own or a third party’s money) to subsidize the exchange. For $B = \sum_{i=1}^n P_i$, $B < 0$ ($B > 0$) means that, after fulfilling all payments, the clearinghouse makes profit (runs a deficit). A mechanism is said to be *incentive compatible* (IC) if $\mathbb{E}[u_s(X(v_s))] \geq \mathbb{E}[u_s(X(v'_s), P(X(v'_s)))]$ for all misreports v'_s .

Strategizing assumptions. We assume the clearinghouse knows that for each agent i and item $j \in H_i \cup W_i$, $v_i(j)$ is contained in a known interval $[\underline{v}_i(j), \overline{v}_i(j)]$. Implicitly, H_i and W_i are known to the clearinghouse. Agent s strategizes over his reports $v'_s(j) \in [\underline{v}_s(j), \overline{v}_s(j)]$ for $j \in H_s \cup W_s$.

Theorem 2 immediately suggests two payment-free mechanisms.

COROLLARY 1. *There exists a mechanism without payments that is WBB, WM, and interim IR.*

COROLLARY 2. *If $\pi_i \geq 0$ for all i , there is a WM, WBB, interim IR payment-free mechanism. By setting $\pi_i \geq 2v_i^*$ for all i , the payment-free mechanism sacrifices WM and becomes fully IR.*

PROOF. Consider the mechanism that begins by solving the corresponding LP-Barter. If the LP is infeasible, then output $X = 0$ i.e., do not reallocate any items. Else, obtain X by rounding the fractional allocation using $\text{RALLOC}(D \cup W \cup E'')$ where D , W , and E are the constraints (4), (6), and (8).

If the LP is infeasible then $\mu_i(X) = 0$, which is interim and ex-post IR. So suppose the LP is feasible with optimal solution x^* . If $\pi_i \geq 0$, interim IR is immediate: $\mathbb{E}[\mu_i(X)] = \mu_i(x^*) \geq \pi_i \geq 0$. If additionally $\pi_i \geq 2v_i^*$, then, by Theorem 1, $\mu_i(X) > \pi_i - 2v_i^* \geq 0$.

Note if LP-Barter is infeasible, then there is no valid fractional reallocation hence the outputted X is WM regardless of the feasibility of LP-Barter. $\square$

In the remainder of this section, we focus on Barter Exchange mechanisms *with* payments, satisfying a subset of WM, WBB, full IR, and IC. We also assume $\pi_i = 0$ for all i unless otherwise stated. Let the mechanism $\neg\text{IC}$ be the mechanism satisfying all four properties except for IC; similarly define $\neg\text{WM}$, $\neg\text{WBB}$, and $\neg\text{full IR}$.

THEOREM 5. *The Barter Exchange mechanisms $\neg\text{WM}$, $\neg\text{IC}$, and $\neg\text{full IR}$ exist.*

PROOF. ($\neg\text{IC}$). Compute an allocation X from x^* as described in Theorem 2 and introduce payments

$$P_i(X) = \mu_i(x^*) - \mu_i(X). \quad (17)$$

These payments allow the clearinghouse to absorb the risk that randomization poses to the agents. Per Theorem 2, X is WM. The mechanism is WBB since $\mathbb{E}[\sum_{i=1}^n P_i] = \sum_{i=1}^n \mathbb{E}[\mu_i(X)] - \mu_i(x) = 0$. Full IR follows because an agent's utility before payments is $\mu_i(X) > -2v_i^*$, per Theorem 2, and after payments it is

$$\mu_i(X) + P_i = \mu_i(x^*) \geq 0. \quad (18)$$

Before proving the next two mechanisms, we introduce some notation. Let

$$\bar{\mu}_i(x) = \sum_{a \in W_i} \bar{v}_i(a)x(a) - \sum_{b \in H_i} \bar{v}_i(b)x(b) \quad (19)$$

$$\underline{\mu}_i(x) = \sum_{a \in W_i} \underline{v}_i(a)x(a) - \sum_{b \in H_i} \underline{v}_i(b)x(b). \quad (20)$$

Analogous to S in (12), let

$$\underline{S} = \{x \in [0, 1]^{|E|} \mid \text{Eq. (4) and } \forall i, \underline{\mu}_i(x) \geq 0\}, \quad (21)$$

$$\bar{S} = \{x \in [0, 1]^{|E|} \mid \text{Eq. (4) and } \forall i, \bar{\mu}_i(x) \geq 0\}. \quad (22)$$

These will be the sets of feasible fractional IR reallocations. Note that these sets are independent of v_s' . Picking reallocations from these sets prevents the problematic scenario where agents report strategically to manipulate the set of feasible fractional reallocations. Since, $\underline{\mu}_i \leq \mu_i \leq \bar{\mu}_i$, then

$$\underline{S} \subseteq S \subseteq \bar{S}. \quad (23)$$

For a strategizing agent s , let

$$x^\dagger = \arg \max_{x \in S} \sum_{i=1, i \neq s}^n \mu_i(x). \quad (24)$$

$\bar{x}^*$ and $\underline{x}^*$ are defined like x^* but maximizing over $\bar{S}$ and $\underline{S}$, respectively; ditto, $\bar{x}^\dagger$ and $\underline{x}^\dagger$.

Define payments

$$P_i(X) = \mu_i(x^*) - \mu_i(X) + \sum_{k \neq i} \mu_k(x^*) - \sum_{k \neq i} \mu_k(x^\dagger) \quad (25)$$

noting that

$$\mathbb{E}[P_i(X)] = \sum_{k \neq i} \mu_k(x^*) - \sum_{k \neq i} \mu_k(x^\dagger). \quad (26)$$

Analogously, define payments $\bar{P}_i(X)$ and $\underline{P}_i(X)$ like $P_i(X)$ but using $(\bar{x}^*, \bar{x}^\dagger)$ and $(\underline{x}^*, \underline{x}^\dagger)$, respectively.

($\neg\text{WM}$). Take $\underline{x}^* = \arg \max_{x \in \underline{S}} \sum_{i=1}^n \mu_i(x)$. Let X be the output of $\text{RALLOC}(\underline{x}^*, D \cup W \cup E'')$ for degree constraints D , edge constraints E'' , and profit constraints

$$W = \{\underline{\mu}_k(x) \geq 0\}_{k=1}^n. \quad (27)$$

We can apply Theorem 1 since all constraints of W are indeed satisfied by $\underline{x}^*$ since $\mu_i(x^*) \geq \mu_i(\underline{x}^*) \geq 0$, and so $\mu(X) > -2v_i^*$. Using payments $\underline{P}_i$, ex-post IR follows because an agent's post-payment utility is

$$\mu_i(X) + \underline{P}_i = \sum_k \mu_k(x^*) - \sum_{k \neq i} \mu_k(\underline{x}^\dagger) \geq 0 \quad (28)$$

where the inequality follows by definition of $\underline{x}^*$ and $\underline{x}^\dagger$ as well as because $\mu_i(x) \geq \underline{\mu}_i(x) \geq 0$ for all $x \in \underline{S}$. Therefore, the mechanism is fully IR. Once more, by definition of $\underline{x}^*$ and $\underline{x}^\dagger$, it is easy to see that $\mathbb{E}[P_i] \leq 0$ for all i ; a sum and the linearity of expectation establishes WBB. Finally, we prove IC. Agent s reports v_s' with the goal of maximizing its expected utility:

$$\mathbb{E}[\mu_s(X) + \underline{P}_s] = \sum_k \mu_k(x^*(v_s')) - \sum_{k \neq s} \mu_k(\underline{x}^\dagger). \quad (29)$$

The right hand sum is independent of v_s' and can be ignored. The left hand sum is precisely the objective, which $x^*(v_s')$ maximizes when s reports truthfully. Therefore, truthfulness will maximize the utility of s .

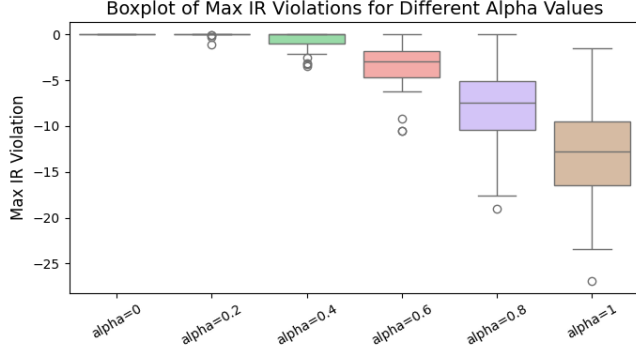
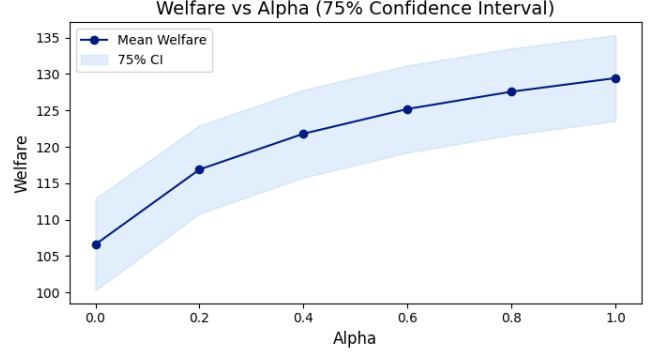
($\neg\text{fully IR}$). Similar to the proof for $\neg\text{WM}$, take $\bar{x}^*$ and round it to X using $\text{RALLOC}(\bar{x}^*, D \cup W \cup E'')$ for

$$W = \{\bar{\mu}_i(x) \geq 0\}_{i=1}^n. \quad (30)$$

Although $\mu_i(\bar{x})$ might be negative, that is okay because this mechanism does not guarantee full IR. Observe that $\sum_{i=1}^n \mu_i(\bar{x}^*) \geq \sum_{i=1}^n \mu_i(x^*)$ because $\bar{S} \supseteq S$. Therefore the mechanism is WM. The proofs for WBB and IC are the same as for $\neg\text{WM}$ but using $\bar{P}_i$. $\square$

6.1 Obstacles towards a WM, fully IR, and IC mechanism

Absent from Theorem 5 is a mechanism satisfying all properties except for WBB. The challenge in developing such a mechanism with our current methods arises from the fact that getting a fractional allocation by maximizing welfare over $\underline{S}$, S , or $\bar{S}$ sacrifices WM, IC, or full IR respectively.

(a) Max IR violations for varying α when requiring $\forall i, \mu_i^\alpha(x) \geq 0$ (b) Welfare for different values of α when requiring $\forall i, \mu_i^\alpha(x) \geq 0$ **Figure 1: Comparison of welfare and maximum IR violations for different values of α when requiring $\forall i, \mu_i^\alpha(x) \geq 0$.**

One way of assuming this problem away would be to strengthen the model of the clearinghouse. If the clearinghouse is “all-having” and can dole out payments including items in H_i , then it suffices to use $\neg$ fully IR as described in the proof of Theorem 5 but increase the payments agents receive with a copy of H_i . Now the mechanism “pays” for these item copies so the mechanism is not WBB, but it becomes fully IR since every item lost by an agent is replaced by the mechanism.

At first glance, this is too strong of an assumption on the clearinghouse. However, this is a realistic assumption when the clearinghouse can either freely copy or procure items for negligibly cheap. An exchange of in-game items hosted by said game’s developers is one such example.

7 Gradually trading WM for IR

Beyond the theoretical guarantees in the previous section, we now explore mechanisms trading off between full IR and WM by taking convex combinations of $\underline{\mu}_i$ and $\overline{\mu}_i$. As seen in the proof of Theorem 5 (particularly, Eqs. (21) and (22)), using constraints “ $\underline{\mu}_i(x) \geq 0$ ” and “ $\overline{\mu}_i(x) \geq 0$ ” gives rise to the mechanisms $\neg$ fully IR and $\neg$ WM. By instead letting

$$\mu_i^\alpha(x) = (1 - \alpha)\underline{\mu}_i(x) + \alpha\overline{\mu}_i(x) \quad (31)$$

and using constraints “ $\mu_i^\alpha(x) \geq 0$ ” for $0 \leq \alpha \leq 1$, we can gradually go from $\neg$ WM (at $\alpha = 0$) to $\neg$ fully IR (at $\alpha = 1$). Define payments $P_i^\alpha(X)$ like in Eq. (25) but using $x^{*,\alpha}$ and $x^{\dagger,\alpha}$, which are in turn defined like x^* and $x^\dagger$ in Theorem 2 and Eq. (24) but instead maximizing over

$$S^\alpha = \{x \in [0, 1]^{|E|} \mid \text{Eq. (4) and } \forall i, \mu_i^\alpha(x) \geq 0\}. \quad (32)$$

Clearly $\underline{S} = S^0 \subseteq S^{\alpha_1} \subseteq S^{\alpha_2} \subseteq S^1 = \overline{S}$ for $\alpha_1 \leq \alpha_2$.

Synthetic data. Experiments are conducted using Python 3.11 and LPs are solved using Gurobi 12. We generate 50 synthetic BAV instances, each with $n = 10$ agents and $m = |\mathcal{I}| = 20$ items. To avoid degenerate instances, for each agent i , $|H_i| \geq 1$ and $|W_i| \geq 1$. First H_i and W_i receive one item each sampled u.a.r.. Next, sample an integer c u.a.r. from $[0, m]$. If $c > 2$, then $c - 2$ items are sampled u.a.r. from $\mathcal{I} \setminus (H_i \cup W_i)$ and then assigned independently and u.a.r. to one

of H_i or W_i so that in the end $|H_i \cup W_i| = \max\{2, c\}$. For each agent i and item j , $v_i(j)$ is an integer sampled u.a.r. from $[10, 19]$. Then $\overline{v_i(j)}$ is sampled u.a.r. from $\{v_i(j) + 2, v_i(j) + 3, \dots, v_i(j) + 7\}$. The true valuation $v_i(j)$ is an integer sampled u.a.r. from $[v_i(j), \overline{v_i(j)}]$. All LPs are solved in less than one second on a commodity laptop.

To compare the trade off between these two mechanisms we will refer to the maximum post-payment loss any agents suffers in a given BAV instance as the *max IR violation*. Here, an agent’s post-payment utility is defined like in Eq. (28) but using payments $P_i^\alpha(X)$. Then the max IR violation is defined to be

$$\min\{0, \min_i u_i(X, P)\}. \quad (33)$$

As expected, our results in Figs. 1a and 1b show that as α increases, so do the welfare and max IR violations. Following the proof of Theorem 5, both of these mechanisms are IC as long as α is fixed independently of reported valuations. Since the choice of randomized payments “absorbs” the randomness to $u_i(X, P(X))$ introduced by X , we can plot the exact post-payment utilities. Thus, the plot’s solid line is the mean post-payment utility over the 50 generated BAV instances. An algorithm designer may then select α appropriate to the underlying application and risk appetite of the participating agents.

We repeat the experiment, but instead for each item j we sample an integer $\underline{v}(j) \in [10, 20]$ and then $\overline{v}(j) = \underline{v}(j) + d$ where d is an integer sampled u.a.r. from $[2, 8]$. Then for each agent i , $v_i(j)$ is known to the clearinghouse to be contained in the interval $[\underline{v}(j), \overline{v}(j)]$. The true valuation $v_i(j)$ is an integer sampled u.a.r. from $[\underline{v}(j), \overline{v}(j)]$. As we can see in Fig. 5, at $\alpha = 0$ (where the mechanism $\neg$ WM takes place), the welfare is 0. In other words, the reallocation that guarantees IR is the empty one. A moment’s reflection can explain this phenomenon. The utility function μ_i^α is “maximally conservative” in that it assumes, for agent i , every item they own is maximally valuable and every item they want is minimally valuable. After assuming this, and that agents have identical valuation intervals, it follows that every trade appears unprofitable to the mechanism; and, therefore, that the most efficient item allocation is the initial one. For sufficiently large α (i.e., as the aforementioned “conservative” assumption is relaxed and

the underlying mechanism transitions into $\neg\text{IR}$), then some trades become profitable and non-empty reallocations are allowed.

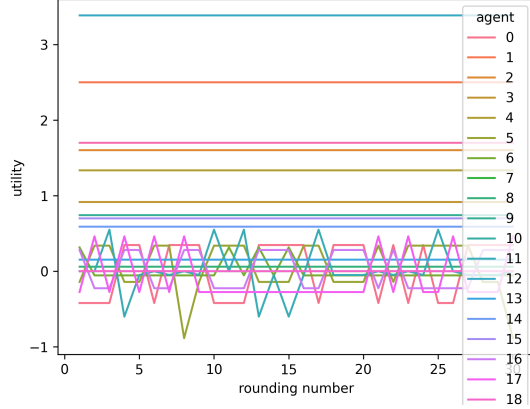


Figure 2: The utility of each agent is plotted in a different color. For many agents, their allocation is a straight line i.e., the edges incident on their items' vertices are all integral in the fractional solution.

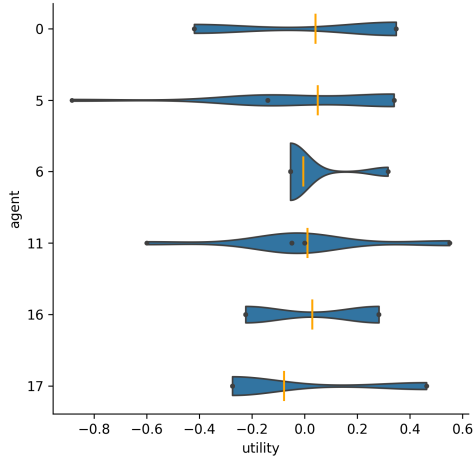


Figure 3: Focusing only on the agents whose allocations included fractional edges, we can observe how their utilities vary. Moreover, for every agent their mean utilities are above or just slightly below zero, per our interim IR guarantee.

8 Conclusion

We introduce and study BAV, a novel centralized barter exchange model motivated by the emergence of markets in online platforms, such as social media and video game markets. This highly applicable model is the most natural extension of a previously studied centralized barter exchange [12]. Alongside the introduction and motivation of this problem, we lay its theoretical foundation

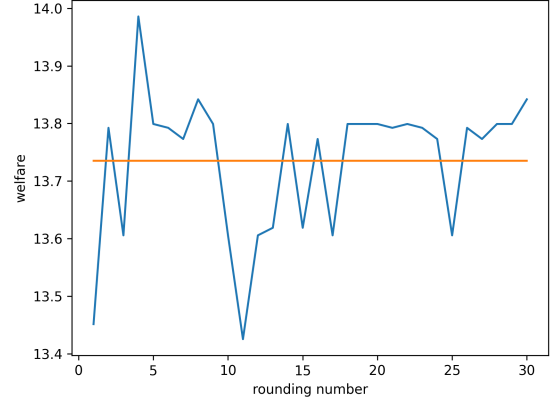


Figure 4: In the 30 fractional solution roundings X , the welfare lies between 13.4 and 14.0. Across all roundings, the mean welfare is 13.74, shown by the orange line.

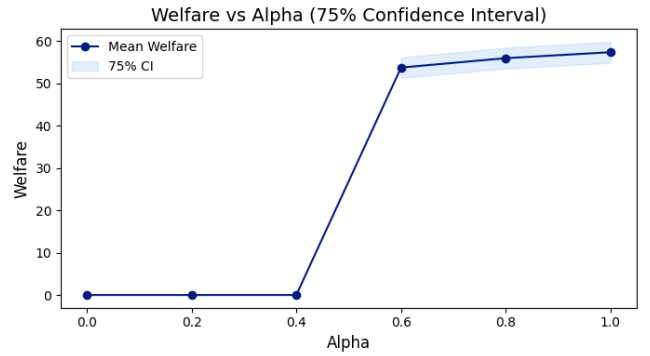


Figure 5: When agents have same valuation intervals for the various items, at $\alpha = 0$, the optimal reallocation is empty. As α increases and trades becomes more profitable, achieving positive welfare becomes possible.

with hardness results, approximation algorithms, and a menu of mechanisms. We leave some interesting questions for future work including the following. First, whether it is possible to improve the guarantee that agents suffer a loss of two items' values. Lastly, is it possible to design a mechanism that is WM, fully IR, WBB, and IC, or can we show these criteria to be incompatible, like in the seminal theorem of Myerson and Satterthwaite [13]. On the practical side, we show that even a simple approach interpolating between two of our mechanisms allows for gradual trade off between welfare and IR. We leave the exploration of additional practical mechanisms for future work.

Acknowledgments

We thank the anonymous reviewers for their helpful feedback about the paper. Sharmila Duppala, Juan Luque, and Aravind Srinivasan were all supported in part by NSF grant CCF-1918749.

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